

COMPASSIQ - AI POWERED TICKET MANAGEMENT SYSTEM



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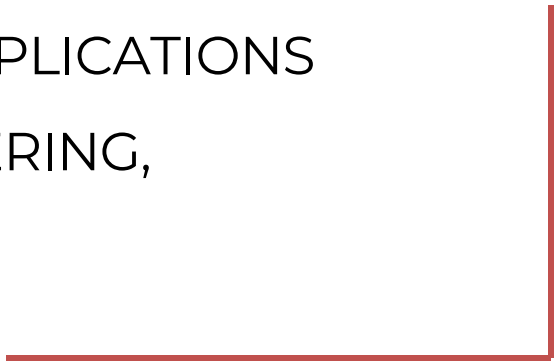


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COMPASSIQ - AI POWERED TICKET MANAGEMENT SYSTEM

- Customer support is one of the most important services offered by product-based companies.
- Every day, organizations receive numerous customer complaints related to technical issues, billing, installation, maintenance, and warranty services.
- Processing these tickets manually is time-consuming and may lead to incorrect ticket routing and delayed responses.



COMPASSIQ - AI POWERED TICKET MANAGEMENT SYSTEM

- CompassIQ is an AI-powered customer support ticket intelligence system designed for a consumer electronics company.
- The application uses Natural Language Processing (NLP) and Machine Learning (ML) to automatically analyze customer complaints, predict the issue category and priority, recommend solutions based on similar historical tickets, and route tickets to the appropriate department.
- The system aims to improve support efficiency, reduce manual effort, and provide a faster and more organized customer support experience.



OBJECTIVES

- To automate customer support ticket management.
- To classify customer complaints into the appropriate department.
- To predict the priority level of each support ticket.
- To route tickets automatically to the correct department.
- To recommend solutions using similar historical support cases.
- To reduce manual effort and response time.
- To improve customer satisfaction through intelligent automation.



EXISTING SYSTEM

- Manual ticket categorization leads to delays and inconsistencies.
 - Incorrect department assignment increases resolution time.
 - Slow response due to manual processing of support requests.
 - High workload for support staff in handling repetitive tasks.
 - No intelligent recommendation for ticket routing or solutions.
 - Reduced customer satisfaction because of delayed and inaccurate support
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PROPOSED SYSTEM

- Accepts customer complaints through an online portal.
- Uses NLP to preprocess complaint text.
- Predicts issue category using Logistic Regression.
- Predicts ticket priority using Machine Learning.
- Retrieves Top-3 similar tickets using TF-IDF and Cosine Similarity.
- Recommends solutions based on previous cases.
- Automatically routes tickets to the appropriate department.
- Provides separate dashboards for Customers, Departments, and Admin.
- Reduces manual effort and improves support efficiency



MOTIVATIONS

- Consumer electronics companies receive a large number of support tickets daily.
- Manual ticket handling is repetitive, time-consuming, and error-prone.
- AI automates ticket classification and priority prediction.
- Historical tickets are used to recommend relevant solutions.
- Improves operational efficiency and reduces manual workload.
- Enables faster and more accurate customer support.



FUNCTIONALITIES

- Customer : Register/Login, submit support tickets, track ticket status, view responses, and provide feedback.
- AI Intelligence : Preprocess complaint text, predict category and priority, retrieve similar tickets, and recommend solutions.



FUNCTIONALITIES

- Department : Login, view assigned tickets, respond to customers, update ticket status, and add resolution notes.
- Administrator : Manage users and departments, monitor tickets, view analytics, track AI performance, and manage the system.

MODULE DESCRIPTION

Customer

- Login/Register
- Submit Support Ticket
- Track Ticket Status
- View Responses
- Provide Feedback & Rating



MODULE DESCRIPTION

AI Intelligence

- Preprocess Complaint Text
- Predict Ticket Category
- Predict Ticket Priority
- Retrieve Similar Tickets
- Recommend Solutions



MODULE DESCRIPTION

Department

- Login
- View Assigned Tickets
- Respond to Customers
- Update Ticket Status
- Add Resolution Notes



MODULE DESCRIPTION

Administrator

- Login
- Manage Users
- View Ticket Statistics
- Monitor AI Predictions
- View Department-wise Reports
- Manage Overall System



DEVELOPING ENVIRONMENT

- Operating System – Windows
- IDE / Code Editor – Visual Studio Code (VS Code)
- Front End – HTML, CSS, JavaScript
- Back End – Python
- Framework – Flask
- Database – MySQL-SQLite
- ML - Scikit-Learn Pandas,Numpy, NLTK ,TF-IDF



THANK YOU